

Project 5: Windows Firewall Lab

My Findings:

Ok, so I opened Windows Defender Firewall with Advanced Security. This comes pre-installed on my PC with a windows 11 OS. I have a few questions that I want to answer:

- What is the difference between inbound and outbound rules?
- What information does a firewall rule contain?
- How do I disable or remove a rule?
- What happened when ICMP was blocked?
- How do I create a rule?
- How can I verify a rule worked?

First thing I see when opening up the dashboard is inbound/outbound rules, an overview of my settings and a handy “Getting Started” section in case someone wasn’t sure what to do. I know that inbound are rules that affect anything trying to connect to me, and outbound rules are me connecting to anything else, so let’s take a look at them.

I went into inbound traffic first and determined that I play way too many games. I was curious why there are some games on here that I don’t have installed anymore and whether having them still on this list would contribute to any slowdown for my PC.

Inbound Rules						
Name	Group	Profile	Enabled	Action	Override	
✓ bg3.exe		Public	Yes	Allow	No	
✓ bg3.exe		Public	Yes	Allow	No	
✓ Bloomtown: A Different Story		All	Yes	Allow	No	
✓ Bloomtown: A Different Story		All	Yes	Allow	No	
✓ Bronzebeards Tavern		Private	Yes	Allow	No	
✓ Bronzebeards Tavern		Private	Yes	Allow	No	
✓ Card Survival: Tropical Island		All	Yes	Allow	No	

It appears that uninstalling the game doesn’t necessarily always uninstall all the rules for it. Also, while it won’t cause significant slowdown, it does pose a minor security risk for there to be unused/needed rules, so let’s see if we can clean them up. Deleting them was a simple right click and delete, but you can also disable them this way.

I am wondering why there are four rules visible for Teams, so I clicked into the properties to get more info. Here it tells me that it is using UDP, can use any port, and where I can find the program file. checked the other three as well to see what was different. I saw that two used UDP and two used TCP but couldn’t find any other differences. Since they seem identical, it’s likely this was caused by an installation or update.

Inbound Rules						
Name	Group	Profile	Enabled	Action	Override	Program
✓ Microsoft 365 Copilot	{78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Prog
✓ Microsoft Teams	{78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Prog
✓ Microsoft Teams	{78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Prog
✓ Microsoft Teams	{78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Prog
✓ Microsoft Teams	{78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Prog

Some rules have different profiles like Public, Private, All, Domain, or even a combination.

Public- This is the most restrictive profile type as it is applied on untrusted networks like hotels or airports. It blocks almost all incoming traffic and prevents the device from being discoverable.

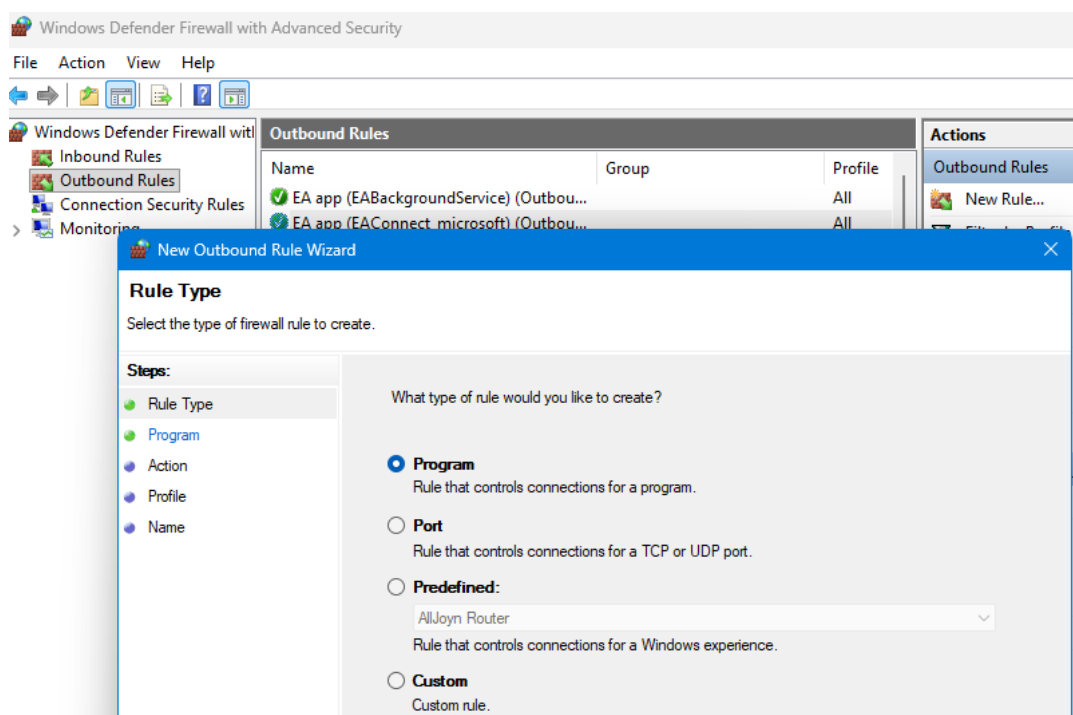
Private- This is for trusted networks like your home or office. This allows you to connect to things like printers and share files on the network.

Domain- This is used when a device connects to an Active Directory.

All- This is used when a rule applies to all of the above profiles.

- All
- Private
- Private
- Public
- Public
- Private
- Public
- Public
- Private
- Domain, Private
- Domain, Private
- Domain, Private
- Private, Public
- Private, Public
- Public

Next, I want to try and make my own rule. I went into Outbound rules and selected the New Rule action on the right. This gave me a few different choices concerning the type of rule I wished to create. I am going to choose Custom so that I can see all the options available to me.



From here I was asked to decide what programs, protocols, and ports this rule would apply to. I chose notepad for the program I was going to test.

Program

Specify the full program path and executable name of the program that this rule matches.

Steps:

- Rule Type
- Program**
- Protocol and Ports
- Scope
- Action
- Profile
- Name

Does this rule apply to all programs or a specific program?

☒ **All programs**
Rule applies to all connections on the computer that match other rule properties.

☐ **This program path:**

Example: c:\path\program.exe
 %ProgramFiles%\browser\browser.exe

Services
Specify which services this rule applies to.

Just to see what options I had and dig around a little I looked at the services option. Selected apply to all and when it got to Protocols, I selected ICMPv4.

Apply this rule as follows:

☒ Apply to all programs and services

☐ Apply to services only

☐ Apply to this service:

Name	Short Name
ActiveX Installer (AxInstSV)	AxInstSV
Agent Activation Runtime_7374b7	AarSvc_7374b7
Aggregated Data Platform Service	ADPSvc
App Readiness	AppReadiness
Application Identity	AppIDSvc
Application Information	Appinfo
Application Layer Gateway Service	ALG
AppX Deployment Service (AppXSVC)	AppXSvc
Auto Time Zone Updater	tzautoupdate

☐ Apply to service with this service short name (example: eventlog):

Protocol type: ICMPv4

Protocol number: Any
Custom
HOPOPT
ICMPv4
IGMP
TCP
UDP
IPv6
IPv6-Route
IPv6-Frag
GRE
ICMPv6
IPv6-NoNxt
IPv6-Opts
VRRP
PGM
L2TP

Local port:

Remote port:

Internet Control Message (ICMP) settings:

I left scope alone, blocked the connection when prompted to choose an action, set to private profile, and named this rule “Test ICMPv4”. To see if this worked, I opened up CMD and tried to ping Google.com.

```
C:\Users\allyr>ping -4 google.com

Pinging google.com [142.250.9.138] with 32 bytes of data:
Reply from 142.250.9.138: bytes=32 time=31ms TTL=107
Reply from 142.250.9.138: bytes=32 time=13ms TTL=107
Reply from 142.250.9.138: bytes=32 time=13ms TTL=107
Reply from 142.250.9.138: bytes=32 time=13ms TTL=107

Ping statistics for 142.250.9.138:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 13ms, Maximum = 31ms, Average = 17ms
```

Weird, that shouldn't have worked..

After troubleshooting, I discovered it only applied to the Private profile while my network was using the Public profile. Once I enabled the Public profile for the rule, ping -4 google.com was successfully blocked.

```
C:\Users\allyr>ping -4 google.com

Pinging google.com [74.125.21.138] with 32 bytes of data:
General failure.
General failure.
General failure.
General failure.

Ping statistics for 74.125.21.138:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```